

RS 232

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INTRODUCTION.

- ❑ **RS-232** (**Recommended Standard 232**).
- ❑ It is basically a interface standards.
- ❑ It is commonly used in computer **serial ports**.
- ❑ The standard defines the electrical characteristics and timing of signals.
- ❑ The current version of the standard is **TIA-232-F**, *issued in 1997.*

History.

- RS-232 was first introduced in 1962.
- For many years, an RS-232-compatible port was a standard feature for serial communication.
- The standard continued to be revised and updated by the Electronic Industries Alliance and since 1988 by the Telecommunications Industry Association (TIA).
- The C revision of the standard was issued in August 1969.



RS-232 Port.

STANDARD.

- ❖ In RS-232, user data is sent as a time-series of bits.
- ❖ Both synchronous and asynchronous transmissions are supported by the standard.
- ❖ RS-232 devices may be classified as Data Terminal Equipment (DTE) or Data Communication Equipment (DCE), this defines that each device which wires will be sending and receiving each signal.
- ❖ DTE refers to terminals and computers that sends and receives data.
- ❖ DCE refers to communication equipment, such as modems, that are responsible for data transferring.

Limitations of the Standard.

- The limited transmission speed, relatively large voltage swing, and large standard connectors motivated development of the universal serial bus(USB) which has displaced RS-232
- Multi-drop connection among more than two devices is not defined.
- Also multi-drop have limitations in speed and compatibility.

DIFFERENT TYPES OF RS-232 CABLES.



RS-232 used in.....

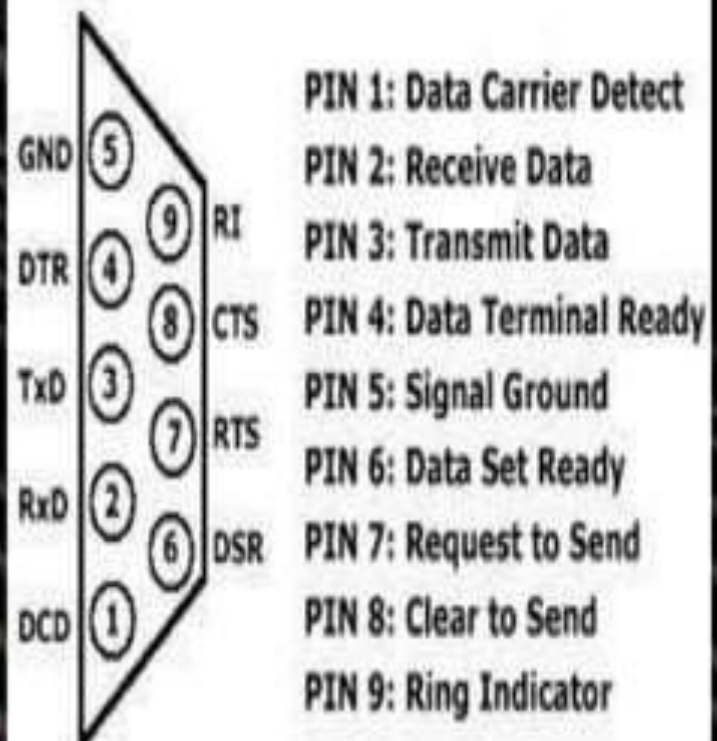


RS-232 pins.

RS232 Pinout on DB25



RS-232 DB-9 Male Pinout



Examining RS-232 handshaking signals.

- ✓ To ensure fast and reliable data transmission between two devices, the data transfer must be coordinated.
- ✓ Many of the pins of RS-232 connector are used for handshaking signals.

PINS OF RS232

1. DTR(Data Terminal Ready)
 2. DSR(Data Set Ready)
 3. RTS(Request to Send)
 4. CTS(Clear to Send)
 5. DCD(Data Carrier Detect)
 6. RI(Ring Indicator)
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Thank you ...